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Chung-Yuan Dye

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Highlights

- We analyze a dynamic advertising problem with psychic stock effect.
- The firm would adopt a static pricing policy over the infinite time horizon.
- The optimal goodwill and psychic stock are both monotonic in the same direction.
- The monotonicity of optimal advertising effort depends on the depreciation rate.

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Optimal joint dynamic pricing, advertising and inventory control model for perishable items with psychic stock effect

Chung-Yuan Dye[a,*]

[a] *Three Mountains Publishing Co., Ltd., Pingtung City, Pingtung County, 90061, Taiwan*

Abstract

In this study, a joint pricing, advertising and inventory control problem will be investigated for a firm selling perishable products with psychic stock effect. The proposed problem is analyzed in the deterministic multi-period setting in which demand at each period depends on not only the amount of inventory displayed and advertising goodwill affected by the firm's current and past advertising efforts but also selling price and freshness index. The objective is to determine the optimal pricing, advertising, and psychic stock strategies maximizing the discount total profit over the infinite planning horizon. Our theoretical results prove that the firm would adopt a static pricing strategy, and then demonstrate that the optimal paths of the advertising goodwill and psychic stock can be determined uniquely. We also show that the convergence of the advertising goodwill towards its equilibrium is from above or below, depending on the relative location of the initial stock of advertising goodwill with respect to the unique equilibrium stock of advertising goodwill. Moreover, a set of structural properties is developed to explore the relationships among the optimal decisions. Finally, the concluding remarks and suggestions will be provided for future studies.

Keywords: pricing, advertising effort, goodwill, psychic stock, freshness index

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1. Introduction

Today, rapid urbanization and rising income levels have increased the consumption of perishable products. As reported by First Research (2013), the US retail grocery industry includes approximately 65000 supermarkets and other grocery stores with combined annual revenues of about $550 billion, including 50% perishable food, 30% non-perishable food and 20% non-food. However, the profitability of individual companies depends on high volume sales and efficient operations due to the lower profit margins. As a result, effective inventory management is critical to the success of the retail industry. Empirical research in marketing has shown that the amount of inventory displayed has a positive effect on sales for many

*Corresponding author
Email address: chungyuandye@gmail.com (Chung-Yuan Dye)

retail products, which therefore acts as a silent salesman. Levin (1972) observed that the presence of inventory has a motivational effect on the customers around it and attracts their impulse buying behaviors in the store. Folwell and Moberg (1993) found that the prices and shelf space of a product contributed greatly to sales and impulse purchasing for the food and beverage retailing. To model the phenomenon in the market, the demand-stimulating role of inventories has been extensively incorporated into theoretical models in the inventory literature. Gupta and Vrat (1986) studied a deterministic inventory system when the demand is dependent on the initial inventory level. Baker and Urban (1988) assumed that demand rate is a power function of the amount of inventory displayed to reflect the diminishing marginal effect of inventory level on demand. Mandal and Phaujdar (1989) took a somewhat different approach, assuming that the demand rate is a linear function of the amount of inventory displayed. Although the linear relation of demand in Mandal and Phaujdar (1989) makes it easier for researchers to analyze the problem than the power relation in Baker and Urban (1988), it is necessary to impose a capacity constraint if building up inventory is profitable. Otherwise, the ordering quantity will reach infinity. Furthermore, it may be more profitable for the firm to maintain higher quantities of the product throughout the replenishment period to sell more items because the displayed inventory promotes sales to customers. Larson and DeMarais (1990) described various categories of inventory and named the inventory that stimulates demand as psychic stock which can be viewed as a minimum stock level. Urban (1992), Chang et al. (2006), Cárdenas-Barrón et al. (2018), and Lee et al. (2017) took into account the psychic stock in the traditional economic order quantity model when the inventories has diminishing returns to the demand rate. Recently, Hsieh and Dye (2017) incorporated the reference price and psychic stock effects into a joint dynamic pricing and inventory control problem for deteriorating items when the demand rate is dependent on the displayed stock level linearly, which belongs to the third type of Goyal and Giri's (2001) classification because they assumed that a constant proportion of physical quantity of items deteriorate over time.

In all the above-mentioned studies works, it did not consider the effect of freshness on the demand of the product. People of the 21st century pay more attention for food safety and quality than the price due to the improvement of living standard. The key feature of fresh-product market is that customers expect the products they buy to be fresh. When planning an inventory management system, the business must be aware that most people think of goods that would expire in a limited period when they hear about perishable items, especially food, beverages, and medicines. Alasalvar et al. (2010) indicated that quality is the key buying cue for seafood purchases. Tsiros and Heilman (2005) and Soethoudt et al. (2012) provided empirical evidences to show that there is a linear negative relationship between shelf life and willingness to pay for the perishable products; and therefore, the consumers would choose the product with the longest expiration date. Taking this into consideration, Fujiwara and Perera (1993) proposed an inventory model for perishable products assuming the product value deteriorates exponentially over time. Sarker et al. (1997) formulated a perishable inventory model and derive the optimal economic order quantity when the age of on-hand stocks has a negative impact on demand. In a closely related work, Bai and Kendall (2008) extended the multi-item EOQ inventory model for perishable items by adding the assumption that the demand rate is increasing in the amount of inventory displayed but

decreasing exponentially in the freshness index of the product. Furthermore, it is well-known that highly perishable items such as fresh fruit, vegetables, dairy and meat products deteriorates continuously over time until reach their expiration dates (i.e., it cannot be sold). Theotokis et al. (2012) found that expiration dates would affect consumers perception on product safety and ultimately change consumer's purchasing decisions. Consequently, Wu et al. (2014) discussed the lot-sizing problem for perishable products with trade credit and freshness dependent demand in which the freshness index is 1 at the beginning of each replenishment period, then gradually declines throughout their lifetime, and finally reaches 0 when the product approaches its expiration date. Teng et al. (2016), Wu et al. (2016), Dobson et al. (2017), Li et al. (2017), Feng et al. (2017), and more recently Li and Teng (2018) then extended the model of Wu et al. (2014) to analyze perishable inventory models with expiration dates under different business scenarios and focused on how to determine the optimal pricing, credit period or replenishment policy.

With businesses rapidly moving into the online environment, advertising acts as a reminder to the existing consumers and attracts new customers as well, which in turn stimulates the actual sales. According to the US Census Bureau, total advertising spending in the US had grown to $170 billion in 2010 (Bureau (2012)). In addition, as reported by Interactive Advertising Bureau (IAB) in 2016, annual Internet advertising revenue in the US alone had grown from $16.9 billion in 2005 to $72.5 billion in 2016. Consequently, advertising is an essential tool for reaching potential consumers and promoting a brand image in business. In the literature of advertising and marketing, the optimal dynamic advertising problems in business and economics fields have been extensively studied over the years. The fundamental result in the development of the advertising model is that of Dorfman and Steiner (1954) who established a joint pricing and advertising strategies for a monopolist. However, the current advertising expenditure in the model of Dorfman and Steiner (1954) does not affect future demand of the product. It has been empirically verified that advertising generates goodwill and aids in building a strong brand image in the market as well as stimulates the actual sales. Nerlove and Arrow (1962) further developed a continuous-time dynamic model to analyze the effect of advertising efforts on the firm's profits, where the advertising efforts are treated as an investment in building the stock of advertising goodwill which affects the present and future demand for the product. Jørgensen and Zaccour (1999) proposed an advertising differential game to a supply chain problem comprising a manufacturer and a retailer by incorporating a convex cost of advertising efforts, indicating increasing marginal costs of the advertising effort.

Furthermore, there are several empirical advertising studies have contradicted the linearity of the advertising response function. Lambin (1976) indicated that the brand advertising has a significant positive impact on the current sales, but the impact of advertising on sales is limited. Simon and Arndt (1980) and Berndt (1991) shown in empirical data and conclude that advertising has diminishing returns to scale. To quantify this phenomenon, Sethi (1975) and Jorgensen (1982) analytically derived the dynamic advertising models under different environmental scenarios when the advertising effort has diminishing returns to the demand rate. Meanwhile, numerous research works, such as Fershtman (1984), Jørgensen and Zaccour (1999), Jørgensen et al. (2001), and Doganoglu and Klapper (2006), constructed

and analyzed different competitive models by assuming that the demand rate is subject to decreasing marginal returns to goodwill. On the other hand, Chintagunta (1993), Luhta and Virtanen (1996), Buratto and Viscolani (2002), Lambertini (2005), and Karray and Martín-Herrán (2009) developed dynamic advertising models when the advertising effort has a decreasing marginal return on the goodwill stock under the Nerlove-Arrow framework. Further, because consumers usually rely on the past prices of products to form a reference price and adjust their purchasing decisions accordingly in a repeated purchasing, Zhang et al. (2013) investigated the impacts of retailer's local advertising and the manufacturer's national advertising on the customers' reference price in a cooperative advertising program. However, the model of Zhang et al. (2013) relies on the assumption that the selling price is a constant. Their work is then extended in a subsequent study by Lu et al. (2016) who considered a joint dynamic pricing and advertising problem for a monopolistic firm by relaxing the assumption of the constant selling price. Unlike the model of Zhang et al. (2013) and Lu et al. (2016) which assume that the reference price is affected by the current sales price and advertising effort directly, Zhang et al. (2015) used different mechanisms and determined various steady-state results when the market demand and customers' reference price both depend on the firm's goodwill stock. Recently, Huang et al. (2012) and Jørgensen and Zaccour (2014) gave comprehensive surveys on the recent trends in the modeling of dynamic advertising. For a detailed survey on the dynamic pricing and advertising models, we refer the readers to Nerlove and Arrow (1962) and the references therein for more details on the subject.

The perishable goods market is growing rapidly due to the global economy and population growth. However, the supermarkets and grocery stores industry is a dynamic and highly competitive industry, and it's getting more and more drastic (First Research (2013), IBIS World Industry Report (2018)). Poor pricing, advertising, and inventory decisions can easily damage the profitability of perishable items in a business. As a result, the grocery retailers should set a competitive selling price to attract customers with an appealing display of stock; use some forms of advertising to establish and reinforce brand awareness, and ultimately, to stimulate revenue. In recent years, there has been growing interest in the use of advertising in revenue management practices, with a focus on integration of dynamic pricing and advertising strategies. While the literature on dynamic pricing and advertising has been largely ignored the impacts of psychic stock and freshness index of a product on the demand, the literature on the joint pricing and inventory control with expiration dates and psychic stock has largely ignored advertising goodwill.To the best of our knowledge, Li and Teng (2018) provides the first analysis of an infinite time horizon dynamic pricing and inventory management for a perishable product with expiration dates and psychic stock, but their focus is on the impact of reference prices on firm's optimal pricing and inventory strategies. Motivated by aforementioned works, we incorporate the effects of pricing, psychic stock, freshness index of the product, and advertising goodwill into a dynamic decision framework for a perishable product over an infinite time horizon where the advertising goodwill affected by the firm's current and past advertising effort. The proposed model belongs to the first type of Goyal and Giri's (2001) classification because we assume the perishable product has a fixed lifetime measured by a number of replenishment periods. This paper contributes to two streams of literature: (1) the infinite planning horizon inventory model for perishable items with price- and stock-dependent demand, and (2) dynamic optimal control

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model in advertising. The remainder of the paper is organized as follows: Section 2 presents the assumptions and notation used throughout this study. In Section 3, the joint pricing, advertising, and inventory control problem is established for a firm, and discrete dynamic programming is used to formulate the multi-period optimization problem and optimize the pricing, advertising and ordering processes. Several structural properties of the optimal decisions and a comparative statics analysis are then investigated by using the properties of the supermodular function to broaden our understanding of the problem. In Section 4, some numerical examples and the results of a sensitivity analysis are discussed to illustrate the features of the proposed model. Finally, the conclusion and the directions for the future research are summarized in Section 5. For the readability of the paper, all the proofs of main results are presented in the online Supplementary Material Part A.

2. Model descriptions

Consider a multi-period profit-maximizing firm who orders and sells a perishable product with a unit cost c over an infinite planning horizon. The firm sets a vector of selling prices over the infinite planning horizon. The selling prices, denoted by $\mathbf{p} = \{p_1, p_2, p_3, ...\}$, are announced upfront, one price for each period where $\underline{p} \leq p_i \leq \overline{p}$, $i = 1, 2, 3, \ldots$. Let h be the holding cost per unit per period. Suppose that the inventory periods are identical with equal length T and the shortages are not allowed. Because the displayed stock level and advertising goodwill may bring about increased sales for the firm and most consumers expect the products they buy to be fresh in the fresh-product market, we assume that demand of the perishable product depends not only on the displayed stocks and advertising goodwill affected by advertising effort but also selling price and freshness index.

At the beginning of each period, the firm need to decide the price, psychic stock, and the advertising effort for that period so as to maximize discounted total profit over the infinite planning horizon. Let G_i denote the stock of advertising goodwill during the ith replenishment period, where $0 \leq G_i \leq \overline{G}$ for $i = 1, 2, 3, \ldots$. Also let $\theta(t)$ denote the freshness index at time t in the ith replenishment period, where $0 \ll \theta(t) \ll 1$ and $\theta'(t) < 0$. Following Sethi (1977), Jørgensen and Zaccour (1999), Bai and Kendall (2008), and Wu et al. (2014), we assume the multiplicative demand function of the perishable product including advertising goodwill, denoted by $D(p_i, G_i, t)$, in the following form:

$$D(p_i, G_i, t) \equiv d(p_i)K(G_i)[I_i(t)]^a\theta(t), \tag{1}$$

where $d(p)$ represents the effect of selling price on demand of the product, $K(G)$ represents the effect of advertising goodwill on demand of the product, $I_i(t)$ is the stock on display at time t, and $0 \leq a < 1$ implies the diminishing marginal effect of inventory level on demand. Also note that the multiplicative structure of demand implies that the price elasticity of demand is independent of the advertising goodwill.

Furthermore, in the literature of inventory theory with price-stock-dependent demand, two types of stock-depend demands have been studied. The first type is that demand rate is proportional to $d(p)I(t)^\beta$ where $0 \leq \beta < 1$, and the second type is that demand rate is

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proportional to $d(p) + \beta I(t)$ where $\beta \geq 0$. Although the linear relation of demand makes it easier to analyze the problem than the power relation, it is necessary to impose a capacity constraint if building up inventory is profitable. Otherwise, the ordering quantity will reach infinity. In this paper, we selected a multiplicative demand function for two reasons. First, it is a commonly used modeling approach for dynamic advertising and perishable inventory models with expiration dates; secondly, it allows us to develop a simple methodology to analyze the relationships among pricing, psychic stock, and advertising efforts.

To formulate the problem more generalized, $d(p)$ is assumed to be non-negative, bounded, continuously differentiable, and decreasing in selling price. Meanwhile, similar to Sethi (1977) and Jørgensen and Zaccour (1999), we assume that $K(G)$ is non-negative, continuous and twice differentiable in G with $K'(G) > 0$ and $K''(G) < 0$, which implies that a greater advertising goodwill may better increase the demand of the product, but the improvement in demand is decreasing as the stock of advertising goodwill increases. Although the investment in advertising increases goodwill, the effect of advertising goodwill on sales is often depreciated over time due to people forgetting. According to Nerlove and Arrow (1962), the stock of advertising goodwill at time τ, denoted by $G(\tau)$, can be described as the following dynamic equation:

$$G'(\tau) \equiv \gamma u(\tau) - \delta G(\tau) \text{ with } G(0) = G_0 \geq 0,$$

where $u(\tau) > 0$ represents advertising effort at time τ, $\delta \in (0, 1)$ represents the depreciation rate of the stock of advertising goodwill, and $\gamma > 0$ represents the advertising efficiency, respectively.

While most dynamic advertising models have been developed assuming a continuous time base, but for intellectual honesty we neglect the forgetting effect of advertising on sales during the period i and assume that the advertising goodwill remains static until the next advertising effort occurs to simplify the analysis of the problem in a multi-period setting, implying that the length of replenishment period is short and the customers do not forget the firm's advertising effort in period i. Therefore, the discrete-time goodwill evolution similar to that in Chintagunta (1993), Luhta and Virtanen (1996), and Karray and Martín-Herrán (2009) can be then represented by the following linear difference equation:

$$G_i = \gamma g(u_i) + (1 - \delta) G_{i-1}, \tag{2}$$

where u_i represents the advertising effort in the period i and the initial stock of the advertising goodwill, G_0, at the beginning of the planning horizon is known and constant. Meanwhile, the advertising investment function g is assumed to be a continuously differentiable, strictly increasing and concave function of u_i to reflect the fact that the advertising effort has a decreasing marginal return on the stock of advertising goodwill. Because adopting advertising effort requires a cost, we denote $T \times f(u)$ by the firm's cost of advertising efforts per replenishment period with an advertising effort u, where $f(u)$ is nonnegative, continuous, strictly increasing and convex in u. Note that $f''(u) > 0$ indicates increasing marginal costs of the advertising effort.

Furthermore, it is difficult to obtain a global explicit estimation of freshness index for a product because many factors (e.g.. packaging method, temperature, refrigeration, etc.)

will affect the freshness of the product. Bai and Kendall (2008) assumed that $\theta(t) = e^{-\sigma t}$ with $\sigma > 0$ to describe the item's freshness index for the perishable product. However, it is well-known that the perishable goods degrade continuously over time until reach their expiration dates. Following Wu et al. (2014), Teng et al. (2016), and more recently Li and Teng (2018), we assume that the freshness index of a perishable product is 1 at the beginning of each replenishment period, then gradually declines over time, and finally reaches 0 when the product approaches its expiration date m, i.e.,

$$\theta(t) = \frac{m-t}{m}, 0 \leq t \leq T \leq m. \tag{3}$$

Further, let $E_i = I_i(T) \geq 0$ be the psychic stock (i.e. ending inventory level) in the ith replenishment period. Tsiros and Heilman (2005) observed that the consumers frequently check the expiration date before making a purchase, and the willingness to pay for a perishable product decreases throughout its shelf life. Theotokis et al. (2012) found that expiration dates would affect consumers perception on product safety and ultimately change consumer's purchasing decisions. Products approaching the use-by date are perceived as products with lower quality by consumers and are therefore less favorable to purchase. A large display of stale products on the shelves might have the opposite effect on sales because this expectation for quality and freshness are the key buying cues for perishable purchases. For easier tractability and clarity of the formulas, we assume in this paper that the psychic stock would be immediately removed from the shelves and be sold at a salvage price s per unit where $c > s > 0$, which implies that the firm can earn revenue by selling the stale items at the end of ith replenishment period at a discounted price and no penalty is imposed.

Given the above notation and assumptions, one can note that the only dynamic component in the model is the stock of advertising goodwill, which is modeled in discrete time with instantaneous impact on the market demand in each period. The pricing and psychic stock are static in each period and evaluated using a deterministic continuous time model. Therefore, we can now model the firm's profit maximizing problem for a perishable product with expiration date. During the ith replenishment period, the depletion of the inventory occurs because of the demand, the inventory level $I_i(t)$ can be described by the following differential equation:

$$\begin{aligned} \frac{\mathrm{d}I_i(t)}{\mathrm{d}t} &= -D(p_i, G_i, t) \\ &= -d(p_i)K(G_i)[I_i(t)]^a \frac{m-t}{m}, \ 0 < t < T. \end{aligned} \tag{4}$$

Solving the differential equation (4) with the boundary condition $I_i(T) = E_i$ yields

$$I_i(t) = \left\{ E_i^{1-a} + (1-a)\frac{t^2 + 2m(T-t) - T^2}{2m} d(p_i)K(G_i) \right\}^{\frac{1}{1-a}}, 0 \leq t \leq T. \tag{5}$$

Utilizing the result of (5), the ordering quantity denoted by Q_i, and the time-weighted inventory during the ith replenishment period can be, respectively, determined as

$$Q_i = I_i(0) = \left[E_i^{1-a} + (1-a)\left(T - \frac{T^2}{2m}\right) d(p_i)K(G_i) \right]^{\frac{1}{1-a}} \tag{6}$$

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and

$$\int_0^T I_i(t)dt = \int_0^T \left\{ E_i^{1-a} + (1-a)\frac{t^2 + 2m(T-t) - T^2}{2m} d(p_i)K(G_i) \right\}^{\frac{1}{1-a}} dt.$$

Hence, the total profit during the ith replenishment period, denoted by $\pi(p_i, E_i, u_i, G_{i-1})$, becomes

$$\begin{aligned}
\pi(p_i, E_i, u_i, G_{i-1}) &= \text{sales revenue} - \text{purchase cost} + \text{salvage value of inventory} \\
&\quad - \text{holding cost} - \text{advertising effort cost} \\
&= p_i\left[Q_i - E_i\right] - cQ_i + sE_i - h\int_0^T I_i(t)dt - Tf(u_i) \\
&= (p_i - c)\left\{ E_i^{1-a} + (1-a)\left(T - \frac{T^2}{2m}\right) d(p_i)K(G_i) \right\}^{\frac{1}{1-a}} - E_i(p_i - s) \\
&\quad - h\int_0^T \left\{ E_i^{1-a} + (1-a)\frac{t^2 + 2m(T-t) - T^2}{2m} d(p_i)K(G_i) \right\}^{\frac{1}{1-a}} dt \\
&\quad - Tf(u_i). \qquad (7)
\end{aligned}$$

Note from (7), we can find that it is not possible to find an explicit analytical solution of $h\int_0^T I_i(t)dt$. Following Bai and Kendall (2008), Wu et al. (2014), Teng et al. (2016), and Li and Teng (2018), the average inventory holding cost, $\frac{hT[Q_i+E_i]}{2}$, is used to approximate the holding cost during the ith replenishment period. After some rearranging, the single period total profit can be rewritten as

$$\begin{aligned}
\pi(p_i, E_i, u_i, G_{i-1}) \approx \left(p_i - c - \frac{hT}{2}\right)\left[E_i^{1-a} + (1-a)\left(T - \frac{T^2}{2m}\right) d(p_i)K(G_i)\right]^{\frac{1}{1-a}} \\
- E_i\left(p_i - s + \frac{hT}{2}\right) - Tf(u_i). \qquad (8)
\end{aligned}$$

Notice from (8) that $p_i - c - \frac{hT}{2}$ represents the gross unit profit. We can observe that

$$\begin{aligned}
&\lim_{E_i\to\infty} \pi(p_i, E_i, u_i, G_{i-1}) \\
&= \lim_{E_i\to\infty}\left(p_i - s + \frac{hT}{2}\right)\frac{\left\{\frac{p_i - c - \frac{hT}{2}}{p_i - s + \frac{hT}{2}}\left[1 + \frac{(1-a)\left(T - \frac{T^2}{2m}\right)d(p_i)K(G_i)}{E_i^{1-a}}\right]^{\frac{1}{1-a}} - 1\right\}}{\frac{1}{E_i}} - Tf(u_i) < 0
\end{aligned}$$

because $c > s$. Without loss of generality, we assume throughout the paper that $\underline{p} - c - \frac{hT}{2} > 0$ and $0 \le E_i < \overline{E} < \infty$ to avoid negative total profit, where $\overline{E}$ is a large number.

We now formulate the firm's pricing and inventory control problem as well as its advertising decision through a multiple period over an infinite horizon. Let

$$\Pi(\mathbf{p}, \mathbf{E}, \mathbf{u}) \equiv \sum_{i=1}^{\infty} z^{i-1}\pi(p_i, E_i, u_i, G_{i-1}) \qquad (9)$$

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denote the firm's discounted total profit over the infinite planning horizon. Consequently, the firm's profit-maximizing problem can be formulated as follows:

$$\max_{(\mathbf{p},\mathbf{E},\mathbf{u})\in\mathbf{\Omega}} \Pi(\mathbf{p},\mathbf{E},\mathbf{u}), \tag{10}$$

where $G_i = \gamma g(u_i) + (1-\delta)G_{i-1}$, $i = 1, 2, 3\ldots$, $\mathbf{p} = \{p_1, p_2, p_3, \ldots\}$, $\mathbf{E} = \{E_1, E_2, E_3, \ldots\}$, $\mathbf{u} = \{u_1, u_2, u_3, \ldots\}$, $z \in (0, 1)$ is a discount factor, and

$$\mathbf{\Omega} \equiv \left\{(\mathbf{p},\mathbf{E}) : \underline{p} \le p_i \le \overline{p}, 0 \le E_i \le \overline{E}, u_i \ge 0, i = 1, 2, 3, \ldots\right\}.$$

3. Structure of the optimal decision strategy

In this section, we will characterize jointly optimal decisions of the firm. Let g^{-1} denote the inverse function of g. The Bellman equation for the standard dynamic programming problem can be written as:

$$V(G) = \max_{\underline{p}\le p\le\overline{p},(1-\delta)G\le y\le\overline{G},0\le E<\overline{E}} J(p, E, y, G), \tag{11}$$

where

$$J(p, E, y, G) \equiv \pi\left(p, E, g^{-1}\left(\frac{y-(1-\delta)G}{\gamma}\right), G\right) + zV(y). \tag{12}$$

If there exists an interior solution, the first-order necessary conditions are

$$\begin{aligned}\frac{\partial J(p,E,y,G)}{\partial p} = &-E + \left[E^{1-a} + (1-a)\left(T - \frac{T^2}{2m}\right)d(p)K(y)\right]^{\frac{1}{1-a}} \\ &+ d'(p)K(y)\left(p - c - \frac{hT}{2}\right)\left(T - \frac{T^2}{2m}\right) \\ &\times\left[E^{1-a} + (1-a)\left(T - \frac{T^2}{2m}\right)d(p)K(y)\right]^{\frac{1}{1-a}-1} = 0,\end{aligned} \tag{13}$$

$$\begin{aligned}\frac{\partial J(p,E,y,G)}{\partial E} = &E^{-a}\left(p - c - \frac{hT}{2}\right)\left[E^{1-a} + (1-a)\left(T - \frac{T^2}{2m}\right)d(p)K(y)\right]^{\frac{1}{1-a}-1} \\ &- \left(p - s + \frac{hT}{2}\right) = 0,\end{aligned} \tag{14}$$

and

$$\begin{aligned}\frac{\partial J(p,E,y,G)}{\partial y} = &d(p)K'(y)\left(p - c - \frac{hT}{2}\right)\left(T - \frac{T^2}{2m}\right) \\ &\times\left[E^{1-a} + (1-a)\left(T - \frac{T^2}{2m}\right)d(p)K(y)\right]^{\frac{1}{1-a}-1}\end{aligned}$$

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$$-\frac{Tf'\left(g^{-1}\left(\frac{y-(1-\delta)G}{\gamma}\right)\right)}{\gamma g'\left(g^{-1}\left(\frac{y-(1-\delta)G}{\gamma}\right)\right)}+zV'(y)=0, \tag{15}$$

respectively. We then characterize the behavior of the firm's optimal pricing strategy for an interior solution in the following theorem. All the proofs of the main results are presented in the online Supplementary Material Part A.

Theorem 1. *The firm would adopt a static pricing strategy over the infinite time horizon, and the optimal princng strategy can be determined from the following equation if an interior solution exists:*

$$\frac{\left(\frac{p-s+\frac{hT}{2}}{p-c-\frac{hT}{2}}\right)^{\frac{1}{a}}-1}{\left(\frac{p-s+\frac{hT}{2}}{p-c-\frac{hT}{2}}\right)^{\frac{1}{a}-1}-1}+\frac{d'(p)}{d(p)}\frac{p-s+\frac{hT}{2}}{1-a}=0. \tag{16}$$

Furthermore, the optimal pricing strategy is unique if $\frac{1}{d(p)}$ is convex in p.

Theorem 1 indicates that the optimal pricing strategy does not depend on the stock of advertising goodwill and psychic stock, and can be uniquely determined because the multiplicative structure of demand implies that the price elasticity of demand is independent of the advertising goodwill. The firm would adopt a static pricing strategy over the infinite time horizon. Comparing the model with Li and Teng (2018) in which the optimal selling price and psychic stock are strategically complementary and converge monotonically to a unique equilibrium, our Theorem 1 happens because we did not consider the impact of reference price on future selling price. Furthermoe, note that the convexity of $\frac{1}{d(p)}$ in Theorem 1 implies that $d''(p) < \frac{d'(p)^2}{d(p)}$. Let $\eta = \frac{d(p)d''(p)}{d'(p)^2}$ denote the measures of the curvature of $d(p)$. It is straightforward to see that $\eta < 1$ because $\frac{1}{d(p)}$ is convex in p indicating that the curvatures of $d(p)$ should not be highly positive, that is, $d(p)$ is not too convex to the origin. Therefore, the convexity of $\frac{1}{d(p)}$ provides a weaker condition than concavity of $d(p)$ with respect to p to ensure the existence and uniqueness of optimal selling price to maximize the discounted total profit.

Let p_e denote the firm's optimal selling price. With the help of Theorem 1, the problem in (11) can be reduced to

$$V(G)=\max_{(1-\delta)G\le y\le \overline{G}, 0\le E\le \overline{E}} J(p_e,E,y,G). \tag{17}$$

Our next focus is the characterizations of the optimal stock of advertising goodwill and psychic stock policies. Let $\{G_i^*\}_{i=1}^{\infty}$ and $\{E_i^*\}_{i=1}^{\infty}$, respectively, denote the optimal sequence of advertising goodwill and psychic sotck over the planning horizon. The following results first present some structural properties of the optimal profit-to-go function given in (17), and then identify the impacts of the initial stock of advertising goodwill on the optimal decisions.

Theorem 2. *For the problem in (17), $V(G)$ is strictly increasing and concave in G; and a unique optimal $(\mathbf{E}^*,\mathbf{u}^*)$ exists such that (17) is maximized if $a\frac{K'(y)}{K(y)}+(1-a)\frac{K''(y)}{K'(y)}\le 0$.*

Proposition 1. *For the problem in (17), we have:*

1. *The optimal value of* $\Pi(p_e, \mathbf{E}, \mathbf{u})$ *increases as* G_0 *increases.*
2. $\{G_i^*\}_{i=1}^{\infty}$ *and* $\{E_i^*\}_{i=1}^{\infty}$ *both increase as* G_0 *increases.*

Theorem 2 indicates that the optimal advertising effort and psychic stock strategies for the problem (17) can be uniquely determined. This, together with Theorem 1, guarantees a unique solution for the problem in (10). Furthermore, Proposition 1 part (1) shows that the optimal value of $\Pi(p_e, \mathbf{E}, \mathbf{u})$ increases as G_0 increases because the higher G_0 means the firm becomes more attractive and this leads to higher demand. One can also observe through the demand function that an increase in the stock of advertising goodwill or psychic stock of a product strengthens that product's demand. Consequently, Proposition 1 part (2) shows that the optimal $\{G_i^*\}_{i=1}^{\infty}$ and $\{E_i^*\}_{i=1}^{\infty}$ both increases as G_0 increases because the benefits accrue from higher current and future stocks of advertising goodwill and psychic stocks.

Having characterized the impacts of initial goodwill on the optimal decisions, we now turn our attention to the properties of the optimal decisions for the firm. Let $\{u_i^*\}_{i=1}^{\infty}$ and $\{Q_i^*\}_{i=1}^{\infty}$, respectively, denote the optimal sequence of advertising efforts and ordering quantity over the planning horizon. The below result summarizes the properties of the optimal decisions as well as relations among them.

Proposition 2. $\{G_i^*\}_{i=1}^{\infty}$, $\{E_i^*\}_{i=1}^{\infty}$, *and* $\{Q_i^*\}_{i=1}^{\infty}$ *are both monotonic in the same direction. However,* $\{u_i^*\}_{i=1}^{\infty}$ *is monotonic in the same direction with respect to* $\{G_i^*\}_{i=1}^{\infty}$ *if* $\frac{dy^*}{dG} \geq 1-\delta$ *for all* $G \in \left[0, \overline{G}\right]$*, but is monotonic in the opposite direction with respect to* $\{G_i^*\}_{i=1}^{\infty}$ *if* $\frac{dy^*}{dG} < 1-\delta$ *for all* $G \in \left[0, \overline{G}\right]$.

The main highlight of Proposition 2 is that $\{G_i^*\}_{i=1}^{\infty}$, $\{E_i^*\}_{i=1}^{\infty}$, and $\{Q_i^*\}_{i=1}^{\infty}$ are both monotonic in the same direction. This happens because a higher psychic stock increases the relevant inventory costs (e.g., holding cost and purchase cost), which in turn reduces the total profit. Consequently, the firm would raise its stock of advertising goodwill to stimulate the demand that provides the best cost-benefit balance and leads to a higher ordering quantity. We can also note that $\{G_i^*\}_{i=1}^{\infty}$ and $\{E_i^*\}_{i=1}^{\infty}$ must respectively converge to their corresponding long-term equilibriums as i approaches infinity because $\{G_i^*\}_{i=1}^{\infty}$ and $\{E_i^*\}_{i=1}^{\infty}$ are both monotonic and bounded. From (6), it is then straightforward to see that $\{Q_i^*\}_{i=1}^{\infty}$ converges to its long-term equilibrium as i approaches infinity. Further, recall that $u_i = g^{-1}\left(\frac{G_i-(1-\delta)G_{i-1}}{\gamma}\right)$, $i = 1, 2, 3, \ldots$. It is clear to see that once $\{G_i^*\}_{i=1}^{\infty}$ converges to its long-term equilibrium, $\{u_i^*\}_{i=1}^{\infty}$ follows, although Proposition 2 guarantees the monotonicity of advertising effort under some specific circumstances. In other words, the effect of current stock of advertising goodwill on the future advertising goodwill only plays a crucial role on determining the optimal advertising effort in the short term. In the long term, however, the optimal advertising effort must converge to its equilibrium.

Let G_e, E_e and u_e respectively denote the equilibrium stock of advertising goodwill, psychic stock, and advertising effort. The firm's equilibrium decisions are given in the following problem.

Proposition 3. *If an interior equilibrium exists, G_e, E_e, and u_e can be simultaneously determined using the following equations:*

$$0 = \left(p_e - c - \frac{hT}{2}\right)\left(T - \frac{T^2}{2m}\right)^{\frac{1}{1-a}} d(p_e)^{\frac{1}{1-a}} K(G_e)^{\frac{a}{1-a}} K'(G_e)$$
$$\times \left\{(1-a)\frac{\left(\frac{p_e - s + \frac{hT}{2}}{p_e - c - \frac{h}{2}}\right)^{\frac{1-a}{a}}}{\left(\frac{p_e - s + \frac{hT}{2}}{p_e - c - \frac{hT}{2}}\right)^{\frac{1-a}{a}} - 1}\right\}^{\frac{a}{1-a}} - [\delta z + (1-z)]\, T \frac{f'\left(g^{-1}\left(\frac{\delta G_e}{\gamma}\right)\right)}{\gamma g'\left(g^{-1}\left(\frac{\delta G_e}{\gamma}\right)\right)}, \tag{18}$$

$$E_e = \left[\frac{(1-a)\left(T - \frac{T^2}{2m}\right) d(p_e) K(G_e)}{\left(\frac{p_e - s + \frac{hT}{2}}{p_e - c - \frac{hT}{2}}\right)^{\frac{1}{a}-1} - 1}\right]^{\frac{1}{1-a}}, \tag{19}$$

and

$$u_e = g^{-1}\left(\frac{\delta G_e}{\gamma}\right). \tag{20}$$

Utilizing the results of (5), it is straightforward to see that the equilibrium ordering quantity, denoted by Q_e, is

$$Q_e = \left[E_e^{1-a} + (1-a)\left(T - \frac{T^2}{2m}\right) d(p_e) K(G_e)\right]^{\frac{1}{1-a}}. \tag{21}$$

With reference to Propositions 2 and 3, The following proposition summarizes the relationships among the initial stock of advertising goodwill and jointly optimal decisions.

Proposition 4. *For the problem in (17), we have:*

(1). If $G_0 < G_e$, then $\{G_i^*\}_{i=1}^{\infty}$, $\{E_i^*\}_{i=1}^{\infty}$, and $\{Q_i^*\}_{i=1}^{\infty}$ all increase monotonically and converge to G_e, E_e, and Q_e, respectively. Furthermore, $\{u_i^*\}_{i=1}^{\infty}$ increases monotonically and converges to u_e if $\frac{dy^*}{dG} \geq 1 - \delta$ for all $G \in \left[0, \overline{G}\right]$, but decreases monotonically and converges to u_e if $\frac{dy^*}{dG} < 1 - \delta$ for all $G \in \left[0, \overline{G}\right]$.

(2). If $G_0 > G_e$, then $\{G_i^*\}_{i=1}^{\infty}$, $\{E_i^*\}_{i=1}^{\infty}$, and $\{Q_i^*\}_{i=1}^{\infty}$ all decrease monotonically and converge to G_e, E_e, and Q_e, respectively. Furthermore, $\{u_i^*\}_{i=1}^{\infty}$ decreases monotonically and converges to u_e if $\frac{dy^*}{dG} \geq 1 - \delta$ for all $G \in \left[0, \overline{G}\right]$, but increases monotonically and converges to u_e if $\frac{dy^*}{dG} < 1 - \delta$ for all $G \in \left[0, \overline{G}\right]$.

(3). If $G_0 = G_e$, then $G_e = G_1^* = G_2^* = G_3^* = \cdots$, $E_e = E_1^* = E_2^* = E_3^* = \cdots$, $Q_e = Q_1^* = Q_2^* = Q_3^* = \cdots$, and $u_e = u_1^* = u_2^* = u_3^* = \cdots$.

Because a higher advertising goodwill attracts more consumers, leading to an increased demand of the product. Proposition 4(1) illustrates that the firm should increase its stock

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of advertising goodwill and psychic stock gradually to stimulate the demand for the product when G_0 is relatively lower, which in turn increases the ordering quantity. Furthermore, we may rewrite $\frac{\mathrm{d}y^*}{\mathrm{d}G} \geq 1-\delta$ as $\delta \geq 1-\frac{\mathrm{d}y^*}{\mathrm{d}G}$, meaning that the depreciation rate of the stock of advertising goodwill is relatively higher which implies that customers forget the past advertisements quickly. Under this circumstance, the firm needs to invest in advertising as much as possible in the beginning to bring the goodwill up rapidly when G_0 is relatively lower. Therefore, the firm should increase its advertising effort gradually and then maintain the advertising goodwill at G_e by spending u_e during the whole planning horizon. Conversely, the firm could invest in advertising at a higher level initially and gradually decrease it to keep the stock of advertising goodwill at a suitable level until it ends up in steady state because the stock of advertising goodwill decays slowly when $\delta < 1-\frac{\mathrm{d}y^*}{\mathrm{d}G}$. Comparing the model with Zhang et al. (2015) and Lu et al. (2016) in which the optimal advertising effort are both monotone decreasing, Proposition 4 happens because the advertising investment function g is assumed to be an increasing concave function of u_i to reflect the fact that the advertising effort has a decreasing marginal return on the stock of advertising goodwill.

On the other hand, Proposition 4(2) demonstrates that the firm should decrease its stock of advertising goodwill and psychic stock gradually to reduce the relevant inventory costs (e.g., holding cost and purchase cost) for the product when G_0 is relatively higher, which in turn increases the ordering quantity. Furthermore, because the customers forgetting past advertising efforts quickly when $\delta > 1-\frac{\mathrm{d}y^*}{\mathrm{d}G}$, the firm could not benefit more from building advertising goodwill. It also shows that the firm should decrease its advertising efforts gradually and let the advertising goodwill decrease to G_e, and then spend u_e to keep it at the equilibrium level. Conversely, the firm is suggested to invest in advertising at a lower level initially and gradually increase it to keep the stock of advertising goodwill at a suitable level until it ends up in steady state if $\delta < 1-\frac{\mathrm{d}y^*}{\mathrm{d}G}$.

In the end of this section, we explore analytically the sensitivity of the equilibrium stock of advertising goodwill, psychic stock and advertising effort with respect to an increase in the relevant model parameters which influence the stock of advertising goodwill. Let $\mathrm{CR}^\delta = a\frac{K'(G_e)}{K(G_e)} + (1-a)\frac{K''(G_e)}{K'(G_e)} + \frac{z\delta(1-a)}{(1-z+z\delta)G_e}$ and $\mathrm{CR}^\gamma = a\frac{K'(G_e)}{K(G_e)} + (1-a)\frac{K''(G_e)}{K'(G_e)} + \frac{1-a}{G_e}$. The details of the representation are given in the following formal proposition.

Proposition 5. *For the problem in (17), if* $a\frac{K'(y)}{K(y)} + (1-a)\frac{K''(y)}{K'(y)} \leq 0$, *we have:*

(1). G_e *and* E_e *both decrease as* δ *increases. Furthermore,* u_e *increases as* δ *increases if* $\mathrm{CR}^\delta \leq 0$*; otherwise,* u_e *decreases as* δ *increases if* $\mathrm{CR}^\delta > 0$.

(2). G_e *and* E_e *both increase as* γ *increases. Furthermore,* u_e *decreases as* γ *increases if* $\mathrm{CR}^\gamma \leq 0$*; otherwise,* u_e *increases as* γ *increases if* $\mathrm{CR}^\gamma > 0$.

Proposition 5.1 first shows that the firm would apply a lower equilibrium stock of advertising goodwill as the goodwill depreciation rate increases, which in turn decreases the equilibrium demand for the product. Consequently, the firm needs to decrease its equilibrium psychic stock to reduce the relevant inventory costs for the product. Furthermore, note

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that the condition $\mathrm{CR}^{\delta} \leq 0$ implies the increased advertising goodwill from investing adverting is larger than the depreciated value of advertising goodwill as δ increases. Under this circumstance, it is benefit for the firm to increase its equilibrium advertising effort to keep the stock of advertising goodwill at a suitable level, which in turn increases the equilibrium demand and leads to the higher discounted total profit. In contrast, the firm should reduce its equilibrium advertising effort as δ increases when $\mathrm{CR}^{\delta} > 0$ because the firm could not gain benefits from a higher depreciated value of goodwill.

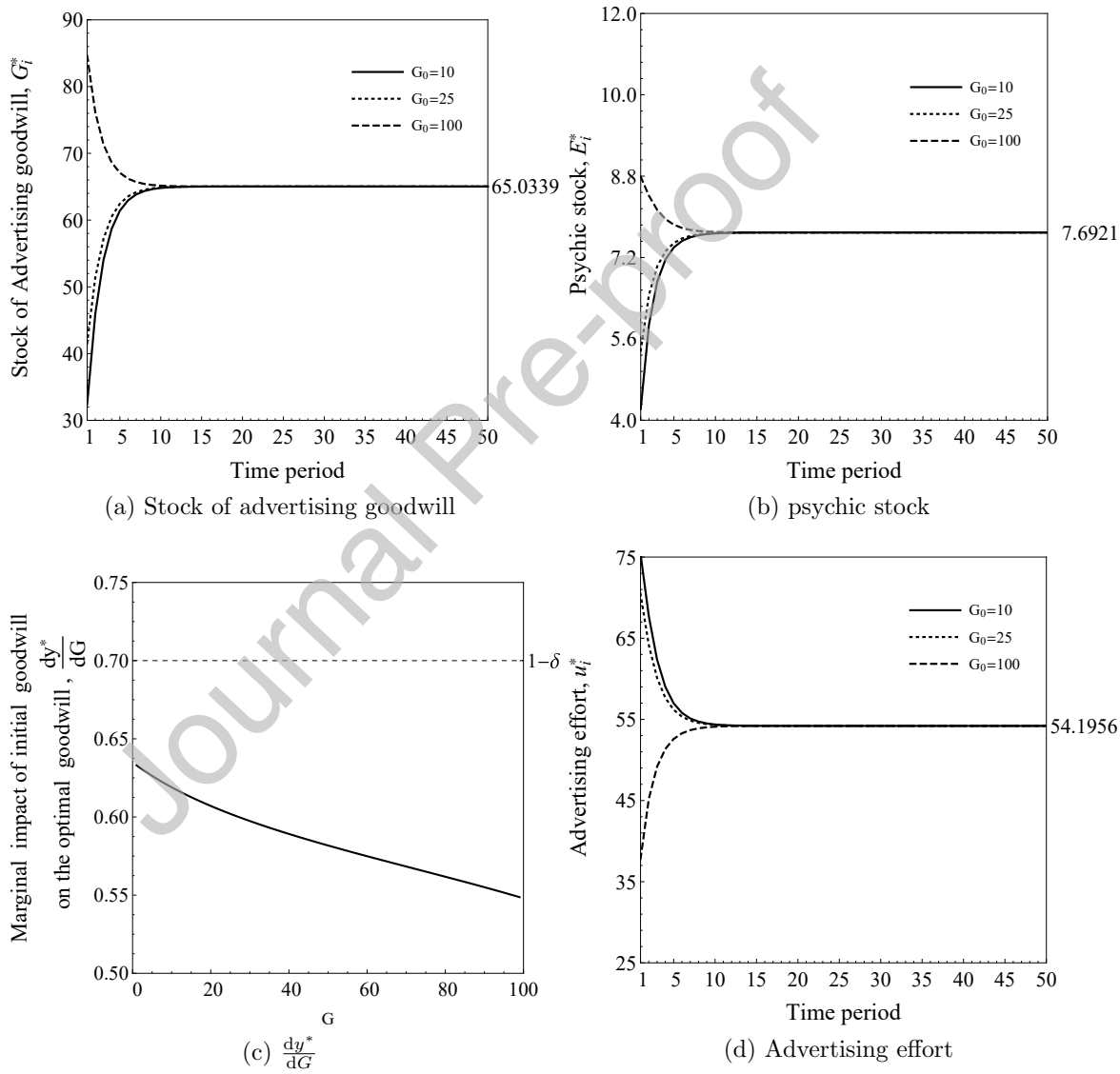
Proposition 5.2 shows that an increase in the advertising efficiency continues to result in an increase in the equilibrium stock of advertising goodwill. This happens because an increase in the advertising efficiency strengthens the marginal contribution of the advertising effort to the stock of advertising goodwill, which in turn increases the equilibrium demand for the product and leads to the higher discounted total profit. As a result, the firm would like to increase its equilibrium psychic stock to pay the higher holding cost and disposal cost. Furthermore, the condition $\mathrm{CR}^{\gamma} \leq 0$ implies the increased advertising goodwill from investing adverting is smaller than the depreciated value of advertising goodwill as γ increases. Therefore, the firm can not benefit more from investing advertising under this circumstance, which in turn decreases its equilibrium advertising effort as γ increases. Conversely, the firm would increase u_e as γ increases if $\mathrm{CR}^{\gamma} > 0$ because a strengthened advertising efficiency causes a higher equilibrium stock of advertising goodwill.

4. Numerical example and sensitivity analysis

In this section we consider groups of numerical examples to illustrate the proposed theoretical results. We select the parameters in the numerical examples as follows: $d(p) = 140 - 4p$, $\underline{p} = 15$, $\overline{p} = 35$, $T = 1$, $c = 15$, $h = 2$, $s = 3$, $a = 0.2$, $m = 3$, $\delta = 0.3$, $\gamma = 0.8$, $z = 0.8$, $v = 1$, $K(G) = \frac{G}{5} - \frac{G^2}{1000}$, and $f(u) = \frac{v \times u^2}{2}$, where $0 \leq G \leq 100$. The solution procedure is implemented using Mathematica Version 11.1 on a personal computer with an AMD Ryzen 7 processor and running Ubuntu 18.04. From Theorem 1, the firm adopt a static pricing policy because the multiplicative structure of demand implies that the price elasticity of demand is independent of the advertising goodwill. The profit maximizing selling price $p_e = 24.5049$ can be determined from using (16). For further demonstrating the effects of advertising efforts on the optimal decisions, we adopt $g(u) = u^{0.8}$ and $\sqrt{u}$, respectively. By setting the different values of G_0, Figures 1 and 2 illustrate the monotonicity properties of $\{G_i^*\}_{i=1}^{\infty}$ and $\{E_i^*\}_{i=1}^{\infty}$ and the convergence of $\{G_i^*\}_{i=1}^{\infty}$, $\{E_i^*\}_{i=1}^{\infty}$, and $\{u_i^*\}_{i=1}^{\infty}$ at each setting of G_0 that we established in Propositions 2 and 3. We also observe from Figures 1 and 2 that $\{G_i^*\}_{i=1}^{\infty}$ and $\{E_i^*\}_{i=1}^{\infty}$ both increase (decrease) monotonically until G_e occurs when the initial stock of advertising goodwill is relatively lower (higher). This happens because a lower (higher) value of G_0 results in a lower (higher) market demand of the product, the firm should increase (decrease) its stock of advertising goodwill and psychic stock gradually to motivate the market demand (to reduce the relevant inventory costs).

Furthermore, for the case of $g(u) = u^{0.8}$, Figure 1(c) shows that $\frac{\mathrm{d}y^*}{\mathrm{dG}}$ is always less than $1 - \delta$, implying that the depreciation rate is relatively lower. Under this circumstance, because the firm can benefit from investing in advertising efforts to bring advertising goodwill

up, it would start with investing in advertising efforts most heavily in the beginning to enhance the advertising goodwill that could generate as much demand as possible, and then gradually decrease the advertising efforts until u_e is reached when G_0 is relatively lower (i.e., $G_0 = 10$). Conversely, when G_0 is relatively higher (i.e., $G_0 = 100$), the firm would invest in advertising efforts at a lower level initially and gradually increase it to keep the stock of advertising goodwill at a suitable level until u_e is reached. This also explains why the $\{u_i^*\}_{i=1}^{\infty}$ in Figure 1(d) increases monotonically in opposite direction as $\{G_i^*\}_{i=1}^{\infty}$ which was indicated in Proposition 4.1.



(a) Stock of advertising goodwill (b) psychic stock

(c) $\frac{dy^*}{dG}$ (d) Advertising effort

Figure 1: The optimal advertising effort and stock of advertising goodwill paths for the dynamic model when $g(u) = u^{0.8}$.

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On the other hand, for the case of $g(u) = \sqrt{u}$, Figure 2(d) demonstrates that $\{u_i^*\}_{i=1}^{\infty}$ is not always monotonic. From Figure 2(c), it shows that $\frac{\mathrm{d}y^*}{\mathrm{dG}} \geq 1 - \delta$ if $G < 21.7114$ and $\frac{\mathrm{d}y^*}{\mathrm{dG}} < 1 - \delta$ if $G > 27.7114$, respectively. Further, we can observe from Figure 2(a) and 2(c) that $\frac{\mathrm{d}y^*}{\mathrm{dG}} > 1 - \delta$ for all $0 \leq G \leq 100$ when $G_0 = 10$, implying that the depreciation rate is relatively higher. Because the customers' forgetting to the firm's past advertisements becomes more quickly as δ is relatively higher, it is difficult to bring the stock of advertising goodwill up to higher levels. The firm is suggested to invest in advertising efforts at a lower level initially and gradually increase it to keep the stock of advertising goodwill at an acceptable level until u_e is achieved. Therefore, $\{u_i^*\}_{i=1}^{\infty}$ for $G_0 = 10$ in Figure 2(d) increases monotonically in same direction as $\{G_i^*\}_{i=1}^{\infty}$. For $G_0 = 50$, however, because $G_i^* \geq 21.7114$ for period $i = 1, 2, 3, 4$ and $G_i^* < 21.7114$ for period $i \geq 5$, $\frac{\mathrm{d}y^*}{\mathrm{dG}} < 1 - \delta$ for period 1, 2, 3, 4 but $\frac{\mathrm{d}y^*}{\mathrm{dG}} \geq 1 - \delta$ for period $i \geq 5$. Consequently, $\{u_i^*\}_{i=1}^{\infty}$ for $G_0 = 50$ in Figure 2(d) first increases monotonically in opposite direction as $\{G_i^*\}_{i=1}^{\infty}$, and then decreases monotonically in same direction as $\{G_i^*\}_{i=1}^{\infty}$ until u_e is reached.

Next, we illustrate Proposition 5 using the same example data as shown in Tables 1 and 2. As δ is relatively higher, the firm needs to reduce its relevant inventory costs and investment in advertisement for the product in the long-term because the customers' forgetting to the firm's past advertising efforts becomes more quickly. Therefore, Table 1 indicates that E_e, G_e, Q_e, and $\Pi(p_e, \mathbf{E}^*, \mathbf{u}^*)$ all decrease as δ increases, as established in Proposition 5.1. Meanwhile, we can observe from Table 1 that u_e first increases as δ is relatively lower because the firm can benefit from investing in advertising efforts, but decreases as δ is relatively higher. On the other hand, Table 2 demonstrates that E_e, G_e, Q_e, and $\Pi(p_e, \mathbf{E}^*, \mathbf{u}^*)$ all increase as γ increases as we shown in Proposition 5.2 because a higher value of γ enhances the impact the advertising effort on the stock of advertising goodwill and market demand of the product. Meanwhile, we can observe from Table 2 that u_e first increases as γ is relatively lower but decreases as γ is relatively higher. This happens because the firm needs to spend a comparatively high amount on advertising to keep the stock of advertising goodwill at a suitable level as γ is relatively higher, bu decrease its investment in advertising efforts to provide the best cost-benefit balance possible as γ is relatively higher.

Moreover, we perform a numerical experimentation to further investigate how the model's equilibrium decisions change with the model parameters. All the computed results of these sensitive analysis are shown in Tables 3 and 4. The numerical results first illustrate that a higher value of c or h results in a higher selling price but a lower equilibrium psychic stock or stock of advertising goodwill, which in turn leads to a decrease in the discounted total profit, equilibrium advertising effort and ordering quantity. This happens because the marginal unit revenue decreases and marginal unit inventory cost increases as c or h increases, the firm should charge a higher selling price to increase the unit marginal profit but decrease its stock of advertising goodwill and psychic stock to reduce the relevant advertising and inventory costs. Meanwhile, Tables 3 and 4 demonstrate that the selling price, equilibrium psychic stock, stock of advertising goodwill, and advertising effort all increase as the salvage value of the product increases. A simple economic interpretation of the results is as follows. As the salvage value increases, the unsold items at the end of the each period can fetch higher revenue. Thereby, the firm could increase the equilibrium stock of advertising goodwill and

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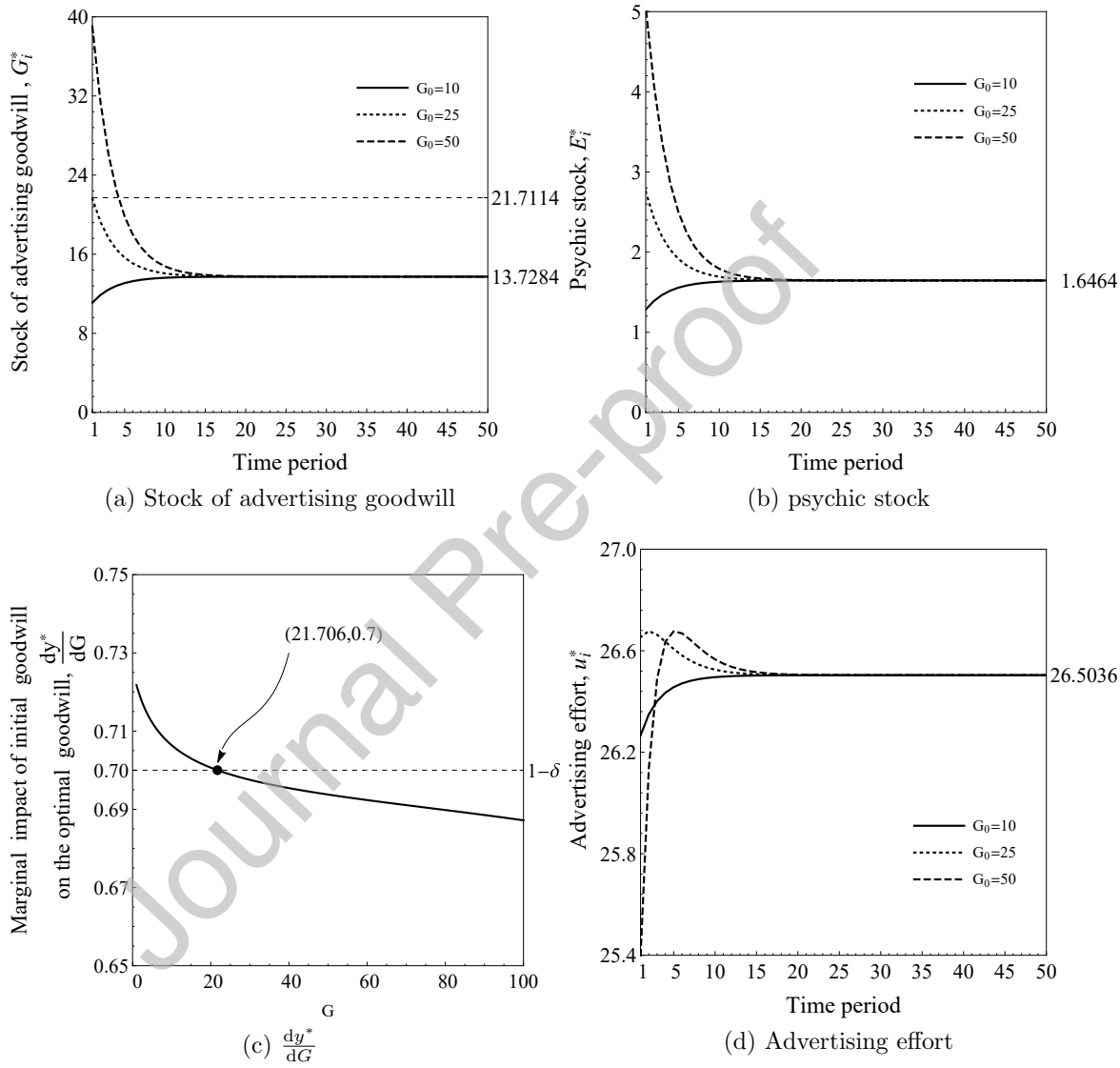

Figure 2: The optimal advertising effort and stock of advertising goodwill paths for the dynamic model when $g(u) = \sqrt{u}$.

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Table 1: Effect of change in δ for the dynamic model.

$g(u)$	δ	p_e	E_e	G_e	u_e	Q_e	Π^*	CR^δ
	0.1	24.5049	8.9971	92.9130	21.4400	1167.2	42 683.2	−0.1101
$u^{0.8}$	0.2	24.5049	8.5778	79.4320	41.9200	1112.8	37 873.9	−0.0336
	0.3	24.5049	7.6921	65.0340	54.1960	997.9	32 100.9	−0.0146
	0.4	24.5049	6.5718	52.4480	59.3440	852.6	26 400.0	−0.0050
	0.5	24.5049	5.4608	42.3220	59.9870	708.4	21 317.9	0.0022
	0.6	24.5049	4.4842	34.4260	58.2010	581.7	17 014.0	0.0088
	0.7	24.5049	3.6751	28.3140	55.2740	476.8	13 453.4	0.0156
	0.8	24.5049	3.0230	23.5670	51.9240	392.2	10 531.5	0.0228
	0.9	24.5049	2.5029	19.8440	48.5280	324.7	8134.5	0.0305
	0.1	24.5049	5.7528	44.8390	8.6239	746.3	30 614.6	−0.0062
$\sqrt{u}$	0.2	24.5049	2.8129	22.0580	8.4506	364.9	21 799.9	0.0138
	0.3	24.5049	1.6464	13.7280	7.7547	213.6	15 705.1	0.0360
	0.4	24.5049	1.0847	9.6195	7.1228	140.7	11 459.9	0.0621
	0.5	24.5049	0.7716	7.2350	6.5940	100.1	8390.9	0.0917
	0.6	24.5049	0.5790	5.7041	6.1526	75.1	6090.6	0.1245
	0.7	24.5049	0.4516	4.6511	5.7804	58.6	4314.2	0.1603
	0.8	24.5049	0.3629	3.8895	5.4622	47.1	2912.0	0.1988
	0.9	24.5049	0.2986	3.3173	5.1870	38.7	1795.7	0.2397

Table 2: Effect of change in γ for the dynamic model.

$g(u)$	γ	p_e	E_e	G_e	u_e	Q_e	Π^*	CR^γ
	0.1	24.5049	0.2098	2.4909	12.3550	27.2	11 524.7	0.3922
$u^{0.8}$	0.2	24.5049	1.0466	9.3340	27.0830	135.8	13 412.3	0.0973
	0.3	24.5049	2.3834	18.9930	39.6500	309.2	16 186.9	0.0417
	0.4	24.5049	3.8697	29.7580	48.5100	502.0	19 500.5	0.0210
	0.5	24.5049	5.2197	40.3040	53.6260	677.2	22 980.3	0.0102
	0.6	24.5049	6.3044	49.8600	55.7060	817.9	26 337.8	0.0028
	0.7	24.5049	7.1137	58.1110	55.6360	922.9	29 402.2	−0.0033
	0.8	24.5049	7.6921	65.0340	54.1960	997.9	32 100.9	−0.0090
	0.9	24.5049	8.0961	70.7470	51.9680	1050.3	34 425.5	−0.0148
	0.1	24.5049	0.0439	0.7062	2.5558	5.7	11 023.4	1.4070
$\sqrt{u}$	0.2	24.5049	0.1526	1.9256	3.7656	19.8	11 458.2	0.5101
	0.3	24.5049	0.3134	3.4508	4.7033	40.7	11 996.1	0.2805
	0.4	24.5049	0.5180	5.2048	5.4870	67.2	12 617.2	0.1827
	0.5	24.5049	0.7595	7.1403	6.1636	98.5	13 308.3	0.1304
	0.6	24.5049	1.0318	9.2231	6.7578	133.9	14 059.1	0.0986
	0.7	24.5049	1.3292	11.4260	7.2848	172.4	14 860.6	0.0774
	0.8	24.5049	1.6464	13.7280	7.7547	213.6	15 705.1	0.0625
	0.9	24.5049	1.9788	16.1100	8.1746	256.7	16 585.1	0.0514

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psychic stock to stimulate the market demand of the product and raise the selling price to gain higher profit margin on each product sold, which in turn results in an increase in the discounted total profit, equilibrium advertising effort and ordering quantity.

Finally, Tables 3 and 4 also show that the optimal p_e remains constant as m or v increases because the optimal p_e does not dependent on m or v. However, we can observe that that the equilibrium psychic stock and stock of advertising goodwill both increase as m increases, but decreases as v increases. The potential reasons are that a higher expiration date or maximum shelf life of the product leads to a higher market demand. Consequently, the firm would like to raise its equilibrium psychic stock and keep the equilibrium stock advertising goodwill at a higher level, resulting in an increase in the discounted total profit and equilibrium advertising effort and ordering quantity. On the other hand, keeping the equilibrium stock of advertising goodwill at a higher level becomes costly as the advertising cost coefficient increases. The firm should lower its equilibrium stock of advertising goodwill to provide the best cost-benefit balance, consequently decreasing the equilibrium demand for the product. Under this circumstance, the firm needs to decrease its equilibrium psychic stock to reduce the holding cost and disposal cost for the product. Therefore, the discounted total profit, equilibrium advertising effort and ordering quantity all decreases as v increases.

5. Concluding Remarks

In this paper, we investigated a joint pricing, advertising, and inventory control problem for perishable items in a multi-period setting where the demand rate varies simultaneously with the selling price, freshness index of the product, stock of advertising goodwill, and amount of inventory displayed. To develop a simple methodology to analyze the problem in a discrete framework, we assume that the demand function follows a multiplicative functional form and the advertising goodwill remains static in period i until the next advertising effort occurs. The first assumption implies that the price elasticity of demand is independent of the advertising goodwill. And the second assumption implies that the length of replenishment period is short and the customers do not forget the firm's advertising efforts in period i. Based on the above assumptions, our theoretical results show that the optimal solution for the problem is unique in which the firm would adopt a static pricing strategy; and the optimal stock of advertising goodwill and psychic stock would both converge monotonically to their corresponding equilibriums. The convergence towards the equilibrium is from above or below, depending on the relative location of the initial stock of advertising goodwill with respect to the unique equilibrium stock of advertising goodwill. Furthermore, we find that the monotonicity of the optimal advertising effort depends on the depreciation rate in the short term. If the depreciation rate is relatively lower, the optimal advertising effort is monotonic in the opposite direction with the optimal stock of advertising goodwill because the firm can bring the advertising goodwill up rapidly. On the contrary, the optimal advertising effort is monotonic in the same direction with the optimal stock of advertising goodwill when the depreciation rate is higher because the customers forget the past advertisements quickly. In the long term, however, the optimal advertising effort must converge to its long-term equilibrium whatever it is monotone or non-monotone.

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Table 3: Effects of changes in the model parameters on the optimal discounted total profit and equilibrium decisions when $g(u) = u^{0.8}$.

Parameter		p_e	E_e	G_e	u_e	Q_e	Π^*
c	10.00	21.9275	77.6590	75.1830	64.9670	1566.2	62 520.0
	12.50	23.1858	25.0280	70.5530	60.0040	1253.1	45 626.1
	15.00	24.5049	7.6921	65.0340	54.1960	997.9	32 100.9
	17.50	25.8590	2.1444	58.3990	47.3740	768.2	21 424.2
	20.00	27.2319	0.5103	50.4580	39.4640	555.3	13 302.2
h	1.00	24.2434	11.0270	66.2320	55.4460	1048.8	34 591.4
	1.50	24.3737	9.2062	65.6380	54.8250	1023.1	33 330.1
	2.00	24.5049	7.6921	65.0340	54.1960	997.9	32 100.9
	2.50	24.6369	6.4313	64.4200	53.5570	973.2	30 902.8
	3.00	24.7697	5.3799	63.7960	52.9090	949.0	29 734.9
s	1.50	24.4926	5.5187	65.0100	54.1710	993.2	32 053.6
	2.25	24.4983	6.4963	65.0210	54.1820	995.3	32 075.2
	3.00	24.5049	7.6921	65.0340	54.1960	997.9	32 100.9
	3.75	24.5125	9.1658	65.0490	54.2110	1001.0	32 131.3
	4.50	24.5213	10.9970	65.0670	54.2300	1004.7	32 167.7
a	0.10	25.0006	0.0344	49.6070	38.6340	408.4	14 677.2
	0.15	24.7427	1.0645	56.8700	45.8290	623.9	21 141.2
	0.20	24.5049	7.6921	65.0340	54.1960	997.9	32 100.9
	0.25	24.2994	32.2490	73.6670	63.3330	1680.8	51 842.0
	0.30	24.1265	106.4900	81.9880	72.3980	3001.6	89 709.2
m	1.50	24.5049	5.4606	59.2720	48.2620	708.4	22 581.6
	2.25	24.5049	6.9305	63.2820	52.3770	899.1	28 811.3
	3.00	24.5049	7.6921	65.0340	54.1960	997.9	32 100.9
	3.75	24.5049	8.1565	66.0160	55.2200	1058.2	34 126.4
	4.50	24.5049	8.4691	66.6430	55.8770	1098.7	35 497.5
v	1.00	24.5049	7.6921	65.0340	54.1960	997.9	32 100.9
	2.00	24.5049	6.3631	50.4180	39.4250	825.5	26 540.3
	3.00	24.5049	5.4067	41.8640	31.2490	701.4	23 514.9
	4.00	24.5049	4.7007	36.1190	25.9840	609.8	21 571.4
	5.00	24.5049	4.1604	31.9440	22.2850	539.7	20 202.6

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Table 4: Effects of changes in the parameters when $g(u) = \sqrt{u}$.

Parameter		p_e	E_e	G_e	u_e	Q_e	Π^*
c	10.00	21.9275	18.8960	16.5090	9.7653	381.1	28 819.9
	12.50	23.1858	5.6988	15.1220	8.7508	285.3	21 546.8
	15.00	24.5049	1.6464	13.7280	7.7547	213.6	15 705.1
	17.50	25.8590	0.4352	12.3010	6.7602	155.9	11 020.4
	20.00	27.2319	0.0999	10.8220	5.7598	108.7	7333.6
h	1.00	24.2434	2.3888	14.0120	7.9554	227.2	16 785.3
	1.50	24.3737	1.9823	13.8700	7.8549	220.3	16 238.6
	2.00	24.5049	1.6464	13.7280	7.7547	213.6	15 705.1
	2.50	24.6369	1.3686	13.5870	7.6547	207.1	15 184.3
	3.00	24.7697	1.1383	13.4450	7.5548	200.8	14 675.7
s	1.50	24.4926	1.1810	13.7230	7.7508	212.5	15 684.5
	2.25	24.4983	1.3903	13.7250	7.7526	213.0	15 694.0
	3.00	24.5049	1.6464	13.7280	7.7547	213.6	15 705.1
	3.75	24.5125	1.9622	13.7320	7.7572	214.3	15 718.3
	4.50	24.5213	2.3547	13.7360	7.7602	215.1	15 734.1
a	0.10	25.0006	0.0086	11.3350	6.1031	101.9	8652.0
	0.15	24.7427	0.2436	12.3870	6.8192	142.8	11 417.6
	0.20	24.5049	1.6464	13.7280	7.7547	213.6	15 705.1
	0.25	24.2994	6.6889	15.5040	9.0278	348.6	22 829.8
	0.30	24.1265	22.5810	17.9460	10.8390	636.5	35 710.1
m	1.50	24.5049	1.1147	12.4770	6.8814	144.6	11 534.4
	2.25	24.5049	1.4596	13.3300	7.4741	189.4	14 272.8
	3.00	24.5049	1.6464	13.7280	7.7547	213.6	15 705.1
	3.75	24.5049	1.7630	13.9600	7.9187	228.7	16 583.9
	4.50	24.5049	1.8425	14.1120	8.0262	239.0	17 177.6
v	1.00	24.5049	1.6464	13.7280	7.7547	213.6	15 705.1
	2.00	24.5049	1.2458	10.8150	5.7552	161.6	14 637.4
	3.00	24.5049	1.0543	9.3924	4.8251	136.8	14 120.4
	4.00	24.5049	0.9352	8.4935	4.2550	121.3	13 795.3
	5.00	24.5049	0.8516	7.8538	3.8582	110.5	13 564.7

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We then focus on the impacts of the relevant model parameters which influence the stock of advertising goodwill on the model's equilibrium decisions. Specifically, we find that the equilibrium advertising effort is pseudoconcave in the depreciation rate or advertising efficiency. The insights gained through the numerical studies indicate that the firm is encouraged to invest more in building the advertising goodwill only if the depreciation rate or advertising efficiency is relatively small. From the numerical results, we also perceive the unit purchase cost c, and demand scale parameter a, or expiration date m has large impacts on the model's equilibrium decisions and discounted total profit, especially in the situation where the advertising investment function g is more sensitive to the advertising effort. It is worth noting that the equilibrium selling price remains constant as the maximum shelf life increases implying that the firm cannot improve the long-term profit through long-term price adjustment. Therefore, the firm is suggested to invest more in effective freshness-keeping efforts to preserve the freshness or prolong the product lifespan because the market demand depends on the freshness index of the product. Meanwhile, setting the value for the demand scale parameter a is a long-term decision for the firm. It is important for the firm to accurately forecast the demand function to set up the pricing, advertising, and inventory strategies in the right way and to optimize the profits from sale of the product.

There are several future research directions. First, the analysis in the proposed model is implemented with a multiplicative demand function, the future works could extend the proposed model with a general class of price and advertising dependent demand because the multiplicative structure implies that the price elasticity of demand is independent of the advertising goodwill. Furthermore, the crucial effects such as competition or time-dependent effects are not included in the model. In this context, it might be possible to extend the monopoly model to a duopoly/oligopoly model (e.g., Krishnamoorthy et al. (2010) and Schlosser (2017).) Meanwhile, the overall effectiveness of a supply chain system is directly affected by the relationships among manufacturers, retailers and customers. Future work could investigate the interaction of psychic stock with advertising between the manufacturer and the retailer as a dynamic Stackelberg game whilst the manufacturer acting as the leader. Finally, an efficient inventory management for perishable products has to not only increase profits but reduce product spoilage during shelf life, which is one of critical actions for the sustainability. It is worthwhile extending the analysis to include the effective investment in freshness-keeping efforts in this model.

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References

Alasalvar, C., Grigor, J.M., Ali, Z., 2010. Practical evaluation of fish quality by objective, subjective, and statistical testing. Handbook of seafood quality, safety and health

Journal Pre-proof

applications , 11–28.

Bai, R., Kendall, G., 2008. A Model for Fresh Produce Shelf-Space Allocation and Inventory Management with Freshness-Condition-Dependent Demand. INFORMS Journal on Computing 20, 78–85.

Baker, R.C., Urban, T.L., 1988. A deterministic inventory system with an inventory-level-dependent demand rate. Journal of the Operational Research Society 39, 823–831.

Berndt, E.R., 1991. The practice of econometrics: classic and contemporary. Addison Wesley Publishing Company.

Buratto, A., Viscolani, B., 2002. New product introduction: goodwill, tinge and advertising cost. Mathematical Methods of Operation Research 55, 55–68.

Bureau, U.S.C., 2012. Statistical Abstract of the United States. Technical Report. US Census Bureau, Washington, DC.

Cárdenas-Barrón, L.E., Shaikh, A.A., Tiwari, S., Trevio-Garza, G., 2018. An eoq inventory model with nonlinear stock dependent holding cost, nonlinear stock dependent demand and trade credit. Computers & Industrial Engineering .

Chang, C.T., Goyal, S., Teng, J.T., 2006. On "an EOQ model for perishable items under stock-dependent selling rate and time-dependent partial backlogging" by dye and ouyang. European Journal of Operational Research 174, 923–929.

Chintagunta, P.K., 1993. Investigating the sensitivity of equilibrium profits to advertising dynamics and competitive effects. Management Science 39, 1146–1162.

Dobson, G., Pinker, E.J., Yildiz, O., 2017. An EOQ model for perishable goods with age-dependent demand rate. European Journal of Operational Research 257, 84–88.

Doganoglu, T., Klapper, D., 2006. Goodwill and dynamic advertising strategies. Quantitative Marketing and Economics 4, 5–29.

Dorfman, R., Steiner, P.O., 1954. Optimal Advertising and Optimal Quality. The American Economic Review 44, 826–836.

Feng, L., Chan, Y.L., Cárdenas-Barrón, L.E., 2017. Pricing and lot-sizing polices for perishable goods when the demand depends on selling price, displayed stocks, and expiration date.

Fershtman, C., 1984. Goodwill and market shares in oligopoly. Economica 51, 271–281.

First Research, 2013. Industry Profile: Grocery Stores and Supermarkets. Technical Report. First Research.

IBIS World Industry Repor, 2018. IBISWorld (2018). Supermarkets and Grocery Stores - Australia Market Research Report.. Technical Report. IBIS World Pty Ltd.

Folwell, R.J., Moberg, A.D., 1993. Factors in retail shelf management impacting wine sales. Agribusiness 9, 595–603.

Fujiwara, O., Perera, U., 1993. EOQ models for continuously deteriorating products using linear and exponential penalty costs. European Journal of Operational Research 70, 104–114.

Goyal, S.K., Giri, B., 2001. Recent trends in modeling of deteriorating inventory. European Journal of Operational Research 134, 1–16.

Gupta, R., Vrat, P., 1986. Inventory models for stock-dependent consumption rate. Opsearch 23, 19–24.

Hsieh, T.P., Dye, C.Y., 2017. Optimal dynamic pricing for deteriorating items with reference price effects when inventories stimulate demand. European Journal of Operational Research 262, 136-150.

Huang, J., Leng, M., Liang, L., 2012. Recent developments in dynamic advertising research. European Journal of Operational Research 220, 591–609.

Jorgensen, S., 1982. A Differential Games Solution to a Logarithmic Advertising Model. The Journal of the Operational Research Society 33, 425.

Jørgensen, S., Taboubi, S., Zaccour, G., 2001. Cooperative advertising in a marketing channel. Journal of Optimization Theory and Applications 110, 145–158.

Jørgensen, S., Zaccour, G., 1999. Equilibrium pricing and advertising strategies in a marketing channel. Journal of optimization theory and applications 102, 111–125.

Jørgensen, S., Zaccour, G., 2014. A survey of game-theoretic models of cooperative advertising. European Journal of Operational Research 237, 1–14.

Karray, S., Martín-Herrán, G., 2009. A dynamic model for advertising and pricing competition between national and store brands. European Journal of Operational Research 193, 451–467.

Krishnamoorthy, A., Prasad, A., Sethi, S., 2010. Optimal Pricing and Advertising in a Durable-Good Duopoly. European Journal of Operational Research 200, 486–497.

Lambertini, L., 2005. Advertising in a dynamic spatial monopoly. European Journal of Operational Research 166, 547–556.

Lambin, J.J., 1976. Advertising, competition, and market conduct in oligopoly over time: an econometric investigation in Western European countries. Contributions to economic analysis, North-Holland Pub. Co.

Larson, P.D., DeMarais, R.A., 1990. Psychic stock: An independent variable category of inventory. International Journal of Physical Distribution & Logistics Management 20, 28–34.

Journal Pre-proof

Lee, W., Wang, S.P., Chen, W.C., 2017. Forward and backward stocking policies for a two-level supply chain with consignment stock agreement and stock-dependent demand. European Journal of Operational Research 256, 830 – 840.

Levin, R.I., 1972. Production operations management: contemporary policy for managing operating systems. McGraw-Hill Companies.

Li, R., Chan, Y.L., Chang, C.T., Cárdenas-Barrón, L.E., 2017. Pricing and lot-sizing policies for perishable products with advance-cash-credit payments by a discounted cash-flow analysis. International Journal of Production Economics 193, 578–589.

Li, R., Teng, J.T., 2018. Pricing and lot-sizing decisions for perishable goods when demand depends on selling price, reference price, product freshness, and displayed stocks. European Journal of Operational Research 270, 1099–1108.

Lu, L., Gou, Q., Tang, W., Zhang, J., 2016. Joint pricing and advertising strategy with reference price effect. International Journal of Production Research 54, 5250–5270.

Luhta, I., Virtanen, I., 1996. Non-linear advertising capital model with time delayed feedback between advertising and stock of goodwill. Chaos, Solitons & Fractals 7, 2083–2104.

Mandal, B., Phaujdar, S., 1989. An inventory model for deteriorating items and stock-dependent consumption rate. Journal of the Operational Research Society 40, 483–488.

Nerlove, M., Arrow, K.J., 1962. Optimal Advertising Policy under Dynamic Conditions. Economica 29, 129–142.

Palda, K.S., 1965. The measurement of cumulative advertising effects. The Journal of Business 38, 162–179.

Parsons, L.J., 1976. A rachet model of advertising carryover effects. Journal of Marketing Research 13, 76–79.

Sarker, B.R., Mukherjee, S., Balan, C.V., 1997. An order-level lot size inventory model with inventory-level dependent demand and deterioration. International Journal of Production Economics 48, 227–236.

Schlosser, R., 2017. Stochastic dynamic pricing and advertising in isoelastic oligopoly models. European Journal of Operational Research 259, 1144–1155.

Sethi, S.P., 1975. Optimal Control of a Logarithmic Advertising Model. Journal of the Operational Research Society 26, 317–319.

Sethi, S.P., 1977. Optimal advertising for the Nerlove-Arrow model under a budget constraint. Journal of the Operational Research Society 28, 683–693.

Simon, J.L., Arndt, J., 1980. The shape of the advertising response function. Journal of Advertising Research 20, 11–28.

Journal Pre-proof

Soethoudt, J., van der Sluis, A., Waarts, Y., Tromp, S., 2012. Houdbaarheidsdatum, verspilde moeite? Technical Report. Wageningen UR-Food & Biobased Research.

Stokey, N.L., Lucas, R.E., Prescott, E.C., 1989. Recursive Methods in Economic Dynamics. Harvard University Press.

Teng, J.T., Cárdenas-Barrón, L.E., Chang, H.J., Wu, J., Hu, Y., 2016. Inventory lot-size policies for deteriorating items with expiration dates and advance payments. Applied Mathematical Modelling 40, 8605–8616.

Theotokis, A., Pramatari, K., Tsiros, M., 2012. Effects of expiration date-based pricing on brand image perceptions. Journal of Retailing 88, 72 – 87. Pricing in a Global Marketplace.

Tsiros, M., Heilman, C.M., 2005. The effect of expiration dates and perceived risk on purchasing behavior in grocery store perishable categories. Journal of marketing 69, 114–129.

Urban, T.L., 1992. Deterministic inventory models incorporating marketing decisions. Computers and Industrial Engineering 22, 85–93.

Wu, J., Al-Khateeb, F.B., Teng, J.T., Cárdenas-Barrón, L.E., 2016. Inventory models for deteriorating items with maximum lifetime under downstream partial trade credits to credit-risk customers by discounted cash-flow analysis. International Journal of Production Economics 171, 105–115.

Wu, J., Ouyang, L.Y., Cárdenas-Barrón, L.E., Goyal, S.K., 2014. Optimal credit period and lot size for deteriorating items with expiration dates under two-level trade credit financing. European Journal of Operational Research 237, 898–908.

Zhang, J., Gou, Q., Liang, L., Huang, Z., 2013. Supply chain coordination through cooperative advertising with reference price effect. Omega 41, 345–353.

Zhang, Q., Zhang, J., Tang, W., 2015. A dynamic advertising model with reference price effect. RAIRO-operations Research 49, 669–688.

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